HOLE OPENER

FIXED-DIAMETER HOLE OPENER

ENLARGEMENT

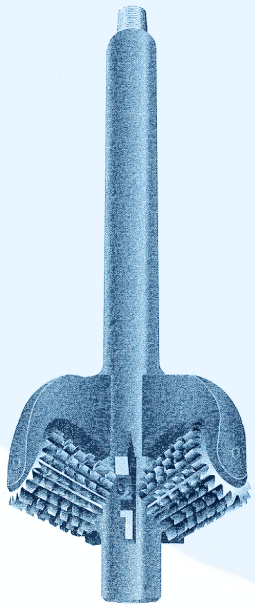
GENERAL DESCRIPTION

FIXED-DIAMETER HOLE OPENER THAT ENLARGES A PREVIOUSLY DRILLED PILOT HOLE FROM THE SURFACE. THE CUTTERS ARE RIGIDLY ATTACHED TO THE BODY: THERE IS NO HYDRAULIC ACTUATION, AND THE TOOL ALWAYS CUTS AT ITS FULL GAUGE. USED MAINLY FOR SURFACE HOLES AND CONDUCTOR CASING, WHERE A BIGGER HOLE GIVES ENOUGH CLEARANCE TO RUN AND CEMENT THE CASING (FOR EXAMPLE, A 17-1/2 IN. HOLE OPENED TO RUN 13-3/8 IN. CASING). THE CHOICE OF CUTTERS LETS IT HANDLE SOFT TO HARD-ABRASIVE FORMATIONS. CUTTER SIZES FROM 26 TO 42 IN.

TECHNICAL CHARACTERISTICS

Std. opening Ø	Min. pilot hole Ø	Fishing neck length	Fishing neck Ø	Top pin API Reg.	Bottom neck length	Bottom neck Ø	Bottom box API Reg.	Body length	Min. bore Ø
26	14	60	9-1/2	6-5/8 - 7-5/8	12	9-1/2	7-5/8	96	1
28	16	60	9-7/8	6-5/8 - 7-5/8	12	9-1/2	7-5/8	96	1
30	18	60	9-7/8	6-5/8 - 7-5/8	12	9-1/2	7-5/8	100	2
32	20	60	10	7-5/8 - 8-5/8	15	9-1/2	8-5/8	100	2
34	22	60	10	7-5/8 - 8-5/8	15	9-1/2	8-5/8	101	2
36	24	60	10	7-5/8 - 8-5/8	15	9-1/2	8-5/8	103	3
38	26	60	10	8-5/8	15	9-1/2	8-5/8	106	3
40	28	60	10	8-5/8	15	9-1/2	8-5/8	106	3
42	30	60	10	8-5/8	15	9-1/2	7-5/8	106	3

Dimensions in inches, approximate. Intermediate sizes or sizes larger than 42 in. available on request.



DESIGN FEATURES

- **Fixed gauge:** cutters rigidly mounted; no hydraulic actuation.
- **Demountable cutters:** easily replaced on the rig floor.
- **Cutters:** sealed-bearing milled tooth or tungsten carbide insert.
- **Long fishing necks:** can be made up below the rotary table despite the large size.
- **Replaceable nozzles:** effective hole cleaning.
- **Wide formation range:** soft to hard-abrasive.

NOZZLES AND CUTTERS

- **Nozzles:** threaded retainer holds the jet nozzle in its sleeve; O-ring seal.
- **Cutter sets:** replacement sets include all parts needed.
- **Rebuild kits:** cutter bearing rebuild kits on special order.

OPERATING PARAMETERS

Formation	Cutter	Weight (lb)	rpm
Soft	Milled tooth	15,000 - 25,000	125 - 150
Medium	Milled tooth	20,000 - 35,000	75 - 85
Hard	Button (TCI)	35,000 - 45,000	50 - 60
Hard, TCI	Extremely hard	30,000 - 45,000	30 - 40

Guideline for 19-3/4 - 26 in. holes (lower sizes need less weight and higher rpm).

GAUGE TOLERANCE (ARMS OPEN)

Size (in)	Plus	Minus
26	+1/8	- 1/16
27 - 42	+5/32	- 1/16
43 and larger	+3/16	- 1/16

Cutter gauge is as dressed, not ground. Applies to milled tooth and TCI.

MASTER DRILLER (CONE ARM VERSION)

Cone-segment arms, replaceable nozzles, bottom box for bit or bullnose. One body takes several arm sizes. Opens up to 36 in.; milled tooth, TCI or PDC cones. Soft to medium-hard formations.

Series	Std. opening Ø	Min. pilot Ø	Body Ø	Wt (lb)
8200	12-1/4	8-1/2	8-1/4	640
9500	12-1/4 / 14-3/4	8-1/2 / 9-1/2	9-1/2	915
11000	17-1/2 / 22-5/8	9-1/2	11-5/8	1,100
15000	17-1/2 / 26-5/8 / 36-5/8	9-1/2 / 12-1/4 / 24-5/8	15-5/8	1,900

HOW TO ORDER